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Agentic systems are revolutionizing problem solving, but deep challenges remain.

- Disambiguated intent
- Guardrails and invariants
- Budget tradeoffs
- Provenance and blast radius analysis
- Necessary HIL escalations



```
service agent: # advanced agent management
# Create with configuration
spawn(task, attachments, context, model_spec)

# Synchronous and Asynchronous Run/Steering.
# Programmatic auto-compaction
compact_run(agent_id, injected_ctx, rounds, tokens)
steer(agent_id, instructions, attachments, wake)
broadcast(agent_id, instruction, attachments)
```

Cost-Aware Graph Executor

```
dag = build_task_graph(spec) # nodes + dependencies

def run_node(n):
    subtask = dag.collect_deps(n) + dag.task(n)
    tier = n.model_spec or "fast" # cost knob

    out = sdk(("agent", "spawn_and_run", {
        "task": subtask,
        "model_spec": tier,
    }))
    dag.mark_done(n, out)

while not dag.done():
    parallel_for(run_node, dag.get_ready())
```

```
# Start of Task: spawn R parallel reviewers
reviewers = [
  sdk("agent", "spawn", {
    "task": review_task,
    "context": "meta",
  })
  for _ in range(R)
]

# Milestone 1: ask R parallel reviews
reviews = parallel_map(
  lambda r: sdk("agent", "compact_run", {
    "agent_id": r.id,
    "injected_context": milestone_ctx,
  }), reviewers,
)

# Milestone 2: reuse same reviewers ...
```

- **Contract-bound solving**
Tradeoffs, invariants, checkpoints, ...
- **Holistic Adaptation**
Scaffolds, models, tools, skills, ...
- **Live flexibility as needed**



```

--- semi-structured sections -----
## Architecture/Vocabulary

## Interface/Usage

--- structured sections -----
'''json STRUCTURED_RELATED_SOURCES

{
  "tressor/ib_service/live.canon.py", // context formation
  "tressor/agent/execution",          // analysis triggers
  "builtin/sdk_libs/self_learning/",  // materialized library
}

'''

'''toml STRUCTURED_CANON

[[canon]]
  kind = "rule"
  oneline = "Compaction must not await smart merge work."
  details = """....."""

[[canon]]
  kind = "technique"
  oneline = "agent.stats(agent_id) is the concise introspection"
  details = """....."""

[[canon]]
  kind = "intent"
  oneline = "Layer 3 self-learning is a user-gated ..."
  details = """....."""

'''

```

```

sdk_libs/self_learning/
|-- SDK_LIB.md
|-- LIB.py
|-- canon_learning.py
|-- optimizer.py
|-- skills/
|   |-- PAST_MERGE_SKILL.md
|   |-- REALIGN_LIVE_CANON_SKILL.md
|   |-- LOCALIZED_AGENTIC_OPTIMIZATION.SKILL.md
|   |-- ...
|-- prompt_templates/
|   |-- CANON_LEARNER_SYSTEM_PROMPT.md
|   |-- OPTIMIZER_SYSTEM_PROMPT.md
|   |-- RAW_PROFILE_MERGE_TASK.md
|   |-- ...
|-- CANON_LEARNER_DENSE_STATE.md # memory

```

*G5 ontologies are much more expressive and UX-friendly
(Provenance, Multi-User Collab, Interactive HTML)*

Tressoir - A Unified SDK And Representation

Interpretable Blueprint

- **System Ontology**
Architecture, relationships, invariants, lessons, ...
More in our G5 system.
- **Materialized Components**
Scaffolds, skills, ...
Offline optimizable.

Human / agent editable for improvement

Composable SDK

- Free-form, advanced orchestration
- Proven component optimization and reuse
- Introspection, metrics

Same SDK for humans and meta agent interpreters

Single, Composable Function for Programmatic Tool Calling

```
sdk(service_name, tool_name, args)
```

- **Shared `sdk(...)` signature**
Unifies builtins, MCPs, plugins, ...
- **Freely composable into complex patterns**
- **Persistent across steps**
- **Available in Python and Lua**
Flexibility / security tradeoffs. Default to Lua.
- **Comprehensive builtin services**
core, web, embed, editor, webview, llm, agent, self-learning

Step N: Parallel Search

```
searches = sdk("core", "parallel_tool_calls", {
    calls = {
        {"web", "search", { query = "agent-native databases" }},
        {"mcp::papers", "search", { query = "HIL agents" }},
    },
})
results = unroll_helper(searches)
print(results)
```

```
relevant_idx = { 2, 5, 8 }

for i = 1, #relevant_idx do
  idx = relevant_idx[i]
  path = sdk("web", "read", {
    url = results[idx].url,
    download_path = "/tmp/sdk-result-" .. idx .. ".pdf",
  })
  sdk("core", "multimodal_view", { path = path })
end
```

Tressoir Is Effective at Designing Agentic Systems

Evaluation Takeaways

- **Effective at offline / hybrid design**
Screenspot-Pro + Bird-Critic
- **Effective at pure online orchestration**
SWE-Bench Pro without learning
- **Caveat: lacking long-term settings**
Weeks- or months-long tasks with periodic human input.
Even hard SWE-Bench-like tasks take ~10-20 minutes.

